

A. PDDL Task Specification

PDDL Domain

- Predicates
- Action Models
- Preconditions and effects

PDDL Problem

- Objects: supports and beam
- Initial state: parts available
- Goal: bridge assembled

Domain defines reusable operators; Problem provides objects, S_0 and g

B. Operator Planner → Linear Reference Plan

Operator Planner
Receives PDDL Domain and PDDL Problem

$$P = (o_1, \dots, o_4)$$

Pick supports

Place supports

Pick beam

Place beam

C. VAL Plan Checking

$$S_0 \xrightarrow{o_1} S_1 \xrightarrow{o_2} S_2 \xrightarrow{o_3} S_3 \xrightarrow{o_4} S_4$$

Valid parameters and action types



Preconditions hold before every action



Final state satisfies the goal: $S_6 \models g$



FAIL?

Return to 2

PASS?

Proceed to 4

D. BuildGraph → Initial Task DAG G

$$G = \text{BuildGraph}(S_0, P, g)$$



Causal edge: condition support between actions



Resource edge: order for resource or action conflicts

START provides conditions already satisfied in S_0

GOAL collects the conditions required by g

